AI RESTORATION STUDIO

# AI-BASED RESTORATION OF DEGRADED SEMICONDUCTOR IMAGES

*End-to-End Dataset Engineering, Deep Restoration Benchmarking, and Offline Desktop Deployment*

| Project element | Verified scope |
|---|---|
| Source domain | NFFA-Europe MEMS scanning electron microscopy |
| Dual benchmark | 2x super-resolution and same-resolution denoising |
| Dataset scale | 4,158 paired samples per task; 8,316 pairs in total |
| Model study | 10 executed Kaggle notebooks; CNN, transformer, multiscale and GAN families |
| Deployment | Restormer -> EDSR offline Windows workstation |

*Prepared for* **Research** *evaluation, academic review, technical archival, and portfolio presentation*

Project team: [Insert team/member names] | Institution: **AI Restoration Studio**

# Document Control and Evidence Basis

This report was reconstructed from the supplied dataset audit document, ten executed Kaggle notebooks, nine packaged model checkpoints, the final PyQt6 application source, its Windows executable and installer, a machine-readable portability audit, and the supplied application screenshot. It does not treat intended notebook behavior as completed evidence unless an executed output or artifact supports the claim.

| Evidence class | Artifacts reviewed | Treatment in this report |
|---|---|---|
| Dataset | Complete technical audit report; Kaggle dataset URL | Counts, geometry, integrity, known limitations |
| Experiments | 5 SR and 5 denoising notebooks with outputs | Reported test tables, curves, configurations, cautions |
| Weights | 9 PyTorch ZIP-format checkpoints | Archive integrity and architecture-key inspection; selective app load evidence |
| Application | app.py, EXE, Inno Setup script and screenshot | Architecture, workflow, packaging, UI and portability |
| External research | Primary model papers | Architecture rationale and technical context |

**EVIDENCE RULE** All benchmark values are transcribed from executed notebook outputs. Because two notebook families use visibly different SSIM implementations or evaluation conventions, PSNR/SSIM values are not assumed to form a perfectly controlled leaderboard.

# Executive Abstract

This project develops a complete semiconductor image-restoration workflow around grayscale MEMS scanning electron microscopy. The work begins with 4,158 real NFFA-Europe source images, constructs two controlled paired-image benchmarks, evaluates complementary convolutional, attention, transformer, multiscale, progressive, and adversarial models, and integrates selected networks into a native offline Windows application. The engineering objective is not generic photo enhancement; it is recovery of fine lines, boundaries, textures, electrodes, cavities, and defect-relevant structures while controlling hallucination risk and computational cost.

SEMICON-SR-2X-v2 isolates spatial reconstruction: every 1024x768 HR target is paired with a 512x384 bicubic LR image. SEMICON-Noise-v2 isolates same-resolution restoration: 1024x768 noisy inputs are paired with clean HR images and span eleven morphology, optics, illumination, intensity, structured charging, and stochastic-noise operators. Both tasks share the same deterministic 70/20/10 source-level split (2,910/831/417), preserving identity consistency across the project.

The dataset audit found all 8,316 task pairs readable, correctly named, geometrically valid, and represented in the manifests. The resulting quality scores were 97/100 for super-resolution, 92/100 for denoising, and 95/100 for Research use. Important cautions remain: 49 exact duplicate HR groups involving 103 files, synthetic-to-real

domain shift, fixed resizing, annotation bars, and stale per-image metadata for five strengthened single-noise groups.

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**READING GUIDE** Sections 3-6 document the data; Sections 7-10 document experiments; Sections 11-12 document deployment; Sections 13-15 convert findings into an actionable **Research** strategy.

# 1. Problem Definition and Industrial Motivation

## 1.1 Image restoration in semiconductor microscopy

SEM imaging is central to the visual inspection of microelectromechanical systems and semiconductor structures. Practical imagery may contain sensor noise, beam-related blur or anisotropy, intensity nonlinearity, illumination falloff, charging artifacts, geometric distortion, and inadequate spatial sampling. These effects can obscure edges and small defects, weaken dimensional interpretation, and reduce downstream reliability. The project therefore frames restoration as a paired inverse problem: estimate a clean, high-resolution representation from a degraded observation while preserving the geometry and contrast patterns that carry inspection value.

Two related but distinct subproblems are isolated. Denoising/restoration maps a 1024x768 degraded image to a clean image of the same dimensions. Super-resolution maps 512x384 to 1024x768 at an exact scale factor of two. Separating these tasks prevents the learning signal for noise removal from being conflated with spatial reconstruction and supports controlled ablations before the models are chained.

## 1.2 Research constraints

- Limited access to the official target-domain data makes external pretraining and disciplined synthetic degradation valuable.
- Kaggle P100/T4 runtimes impose strict VRAM and session-time limits, motivating patch training, AMP, tiled inference, checkpoint resume, and deferred IQA computation.
- PSNR and SSIM reward fidelity, while perceptual metrics and visual inspection are needed to detect oversmoothing or GAN hallucination.
- Inference speed and portability matter because a judging demonstration must be stable, offline, and understandable to non-developers.

## 1.3 Central engineering question

**RESEARCH QUESTION** Can a reproducible external SEM dataset, trained with architecture-appropriate pipelines and delivered through a fixed offline restoration workstation, provide a credible foundation for later target-domain adaptation without sacrificing inspection-relevant structure?

# 2. Project Objectives and Contributions

## 2.1 Objectives

1. Create controlled paired datasets for pure 2x super-resolution and same-resolution multi-degradation restoration.
2. Prevent source-level leakage through deterministic splitting before pair construction.
3. Establish broad CNN, attention, transformer, multiscale, progressive, and GAN benchmarks using common manifests.
4. Evaluate fidelity, structural, perceptual, no-reference, efficiency, and resource metrics.
5. Deploy a fixed Restormer -> EDSR pipeline in a professional native Windows application.
6. Document integrity, limitations, and release risks sufficiently for academic and industrial review.

## 2.2 Delivered contributions

| Contribution | Technical value | Evidence |
|---|---|---|
| Dual paired benchmark | Task isolation with shared identities | 4,158 pairs per task |
| Read-only dataset audit | Geometry, corruption, pairing, duplicate and balance checks | 95/100 Research score |
| Ten model benchmarks | Five SR and five denoising families | Executed notebooks and test tables |
| Resource-aware training | AMP, aligned crops, tiling, checkpointing and resume | Kaggle notebook implementations |
| Offline workstation | Fixed two-stage inference with compact quality indicators | PyQt6 source and executable |
| Portable bundle audit | Embedded weights and native dependencies | 1,540-entry PyInstaller archive audit |

The most important contribution is the end-to-end continuity between dataset identities, training artifacts, checkpoint packaging, and the user-facing demonstration. This continuity makes the project more than a collection of disconnected model experiments.

# 3. Dataset Source and Construction

## 3.1 Source domain

The source imagery is the NFFA-Europe 100% dataset category MEMS_devices_and_electrodes. The supplied analysis reports 4,158 readable grayscale JPEG SEM images. Native dimensions were dominated by 1024x768 imagery, with larger 2048x1536 and 3072x2304 examples and a small number of 1023x767 images. No corrupted source files were found in the original analysis.

## 3.2 Standardization and split

For version 2, every source image was standardized to 1024x768 before task generation. A fixed seed of 2026 established one source-level ledger: 2,910 train images, 831 validation images, and 417 test images. Both task datasets reuse the same identities. This design supports paired cross-task analysis and prevents a source file from moving between splits during model-specific experiments.

| Split | Images | Percent | Noise single | Noise combined |
|---|---|---|---|---|
| Train | 2,910 | 69.99% | 1,455 | 1,455 |
| Validation | 831 | 19.99% | 416 | 415 |
| Test | 417 | 10.03% | 209 | 208 |
| Total | 4,158 | 100% | 2,080 | 2,078 |

## 3.3 Storage organization

SEMICON-SR-2X-v2 stores HR and LR directories beneath train, val, and test. SEMICON-Noise-v2 stores single and combined categories beneath every split, each containing HR and LR directories. The audit reports PNG storage of approximately 2.67 GiB for SR and 4.23 GiB for Noise, totaling 6.90 GiB and 16,632 image files.

**REPRODUCIBILITY STRENGTH** The shared split ledger is a strong design decision. It permits the denoising and SR stages to be evaluated independently or sequentially without silently changing source identities.

# 4. Super-Resolution Benchmark Design

## 4.1 Pair generation

SEMICON-SR-2X-v2 is deliberately simple: the clean standardized image is retained as a 1024x768 grayscale HR target and downsampled using OpenCV INTER_CUBIC to 512x384. No blur, noise, compression, or geometric distortion is introduced. The resulting inverse problem is classical bicubic 2x single-image super-resolution.

| Property | Specification |
|---|---|
| Input | 512x384 grayscale PNG |
| Target | 1024x768 grayscale PNG |
| Scale | Exactly 2x in width and height |
| Degradation | OpenCV bicubic downsampling only |
| Pairs | 4,158 |
| Audit score | 97/100 |

## 4.2 Scientific role

The single-kernel design offers high internal validity: architecture differences can be studied without uncertainty from a mixed degradation model. It is particularly suitable for SRCNN, EDSR, RCAN, SwinIR, and related fidelity-oriented methods. It is not a blind or real-world SR benchmark; performance may fall when official images contain unknown blur, noise, compression, or scanner-specific sampling effects.

## 4.3 Risks

- Standardization can alter native pixel-to-physical-scale relationships.
- A model can learn the fixed bicubic inverse without generalizing to other kernels.
- SEM scale bars and instrument labels may become shortcuts or disproportionately influence metrics.
- Exact source duplicates can make validation/test estimates optimistic if duplicate groups cross splits.

# 5. Denoising Benchmark Design

## 5.1 Single and combined policies

SEMICON-Noise-v2 contains 4,158 clean/noisy pairs at 1024x768. Single samples apply exactly one operator, supporting interpretable operator-specific learning. Combined samples apply all eleven operators sequentially, creating a difficult multi-artifact restoration task. The allocation is nearly balanced: 2,080 single samples and 2,078 combined samples.

## 5.2 Degradation inventory

| Degradation | Mechanism | Effective single range | N |
|---|---|---|---|
| Linewidth/CD bias | Morphological erosion/dilation | radius 1-3 px | 189 |
| Corner rounding | Gaussian edge smoothing | 0.5025-1.9974 px | 185 |
| Beam astigmatism | Anisotropic blur | sigma x/y about 0.30-1.20 | 202 |
| Barrel distortion | Radial geometric warp | k1 0.000294-0.119901 | 189 |
| Pincushion distortion | Inverse radial warp | k1 -0.119977 to -0.000110 | 213 |
| Salt-and-pepper | Impulse noise | p 0.000101-0.009999 | 174 |
| Charging streaks | Structured horizontal bands | intensity 0.10-0.28; 8-20 streaks | 184 |
| Vignetting | Radial illumination falloff | strength 0.22-0.55 | 197 |
| Speckle noise | Multiplicative noise | sigma 0.08-0.20 | 190 |
| Gaussian noise | Additive sensor-like noise | sigma 0.035-0.095 | 169 |
| Gamma adjustment | Nonlinear intensity response | 0.50-0.75 or 1.35-1.70 | 188 |

## 5.3 Metadata provenance issue

**HIGH-PRIORITY LIMITATION** Five single-noise groups were regenerated at stronger settings, but the manifest was intentionally preserved byte-for-byte. The image bytes reflect the stronger ranges above, while the original per-image parameters remain stale unless a repair log or corrected manifest is published.

# 6. Dataset Audit and Training Readiness

## 6.1 Integrity results

| Validation check | SR | Noise | Status |
|---|---|---|---|
| Manifest records | 4,158 | 4,158 | PASS |
| Missing HR | 0 | 0 | PASS |
| Missing LR/noisy | 0 | 0 | PASS |
| Unreadable pairs | 0 | 0 | PASS |
| Filename mismatch | 0 | 0 | PASS |
| Geometry errors | 0 | 0 | PASS |
| Duplicate sample IDs | 0 | 0 | PASS |
| Orphan images | 0 | 0 | PASS |

## 6.2 Sampled image statistics

| Role | Mean | Std | Entropy | Contrast | Sharpness |
|---|---|---|---|---|---|
| SR HR | 0.43641 | 0.23953 | 6.47798 | 0.81854 | 5,353.23 |
| SR LR | 0.43572 | 0.23432 | 6.40958 | 0.81103 | 5,028.22 |
| Noise HR | 0.44148 | 0.24131 | 6.51922 | 0.82596 | 4,976.08 |
| Noise input | 0.42961 | 0.22415 | 6.63810 | 0.73341 | 4,052.79 |

Bicubic downsampling slightly lowers entropy, contrast, and Laplacian sharpness. The noisy set has higher entropy but lower robust contrast and sharpness than clean HR, consistent with mixed stochastic, blur, and intensity artifacts. These are dataset descriptors rather than restoration scores.

## 6.3 Duplicate risk

SHA-256 analysis identified 49 exact duplicate source groups involving 103 HR files. Filename-level splitting prevents direct patch leakage from one filename but does not guarantee that identical content is grouped. The current release remains useful for continuity; a future benchmark revision should group exact and perceptual duplicates before splitting.

| Scorecard | Score | Interpretation |
|---|---|---|
| Super-resolution | 97/100 | Excellent geometry and integrity; limited kernel diversity |
| Denoising | 92/100 | Strong pairing and coverage; metadata repair required |
| Research use | 95/100 | Highly useful external pretraining resource |

# 7. Experimental Protocol and Metrics

## 7.1 Common training design

All notebooks load the published manifest-defined splits rather than creating model-specific partitions. Training uses aligned random crops and paired flips/rotations. Validation and testing preserve identity and apply no random augmentation. AMP reduces GPU memory, gradient clipping controls unstable updates, AdamW is the dominant optimizer, cosine scheduling decays learning rate, and checkpoint state supports interruption/resume on Kaggle.

| Task | Training target | Typical crops | Evaluation |
|---|---|---|---|
| Super-resolution | LR patch -> aligned 2x HR patch | 96-128 LR pixels | Full/tiled 512x384 -> 1024x768 |
| Denoising | Noisy patch -> clean same-size patch | 192-256 pixels | Full/tiled 1024x768 |

## 7.2 Metric portfolio

| Metric | Role | Direction | Important interpretation |
|---|---|---|---|
| PSNR | Pixel fidelity | Higher | Primary checkpoint criterion; sensitive to MSE |
| SSIM | Structural similarity | Higher | Implementation/window choices must match |
| MAE/MSE/RMSE | Absolute/quadratic error | Lower | Useful for transparent error magnitude |
| LPIPS | Learned perceptual distance | Lower | Natural-image backbone; interpret cautiously for SEM |
| NIQE/BRISQUE/PIQE | No-reference quality | Lower | Natural-scene assumptions may not fit SEM |
| FSIM | Feature similarity | Higher | Complements pixel/structure metrics |
| Entropy | Intensity information content | Context | Not intrinsically better when higher |
| Latency/FPS/VRAM/params | Operational cost | Depends | Must state device, resolution, warm-up and tiling |

**METRIC CAVEAT** The supplied notebooks were developed in two code families. Their SSIM and validation conventions are not identical, and some validation LPIPS series are zero placeholders. Final test metrics are still valuable within each notebook, but cross-family ranking should emphasize PSNR, visual evidence, and a future standardized rerun.

# 8. Super-Resolution Models and Results

## 8.1 Implemented model families

| Model | Architecture | Project role |
|---|---|---|
| SRCNN | Shallow three-layer CNN after bicubic pre-upscaling | Fast sanity baseline; limited capacity |
| EDSR | Residual CNN without unnecessary batch-normalization modules | Strong fidelity, stable training, deployable size |
| RCAN | Residual-in-residual network with channel attention | High-frequency focus; heavier than EDSR |
| ESRGAN | RRDB generator with relativistic adversarial training | Perceptual sharpness; hallucination and instability risk |
| SwinIR | Residual Swin Transformer blocks with window attention | Context modeling with tiled, memory-aware inference |

## 8.2 Executed test results

| Model | PSNR dB | SSIM | Parameters | Latency | Approx. FPS |
|---|---|---|---|---|---|
| SRCNN | 26.116 | 0.9697 | 57,281 | 0.0238 s | 42.04 |
| EDSR | 29.355 | 0.9819 | 1,367,553 | 0.1047 s | 9.55 |
| RCAN | 28.918 | 0.7197* | 4,092,297 | 0.4722 s | 2.12 |
| ESRGAN | 25.977 | 0.6859* | 5,978,177 | 0.6840 s | 1.46 |
| SwinIR | 26.392 | 0.7402* | 515,281 | 0.7002 s | 1.43 |

**SSIM values marked with an asterisk originate from a different notebook implementation family and should not be compared directly with the EDSR/SRCNN SSIM values until metric code is standardized.*

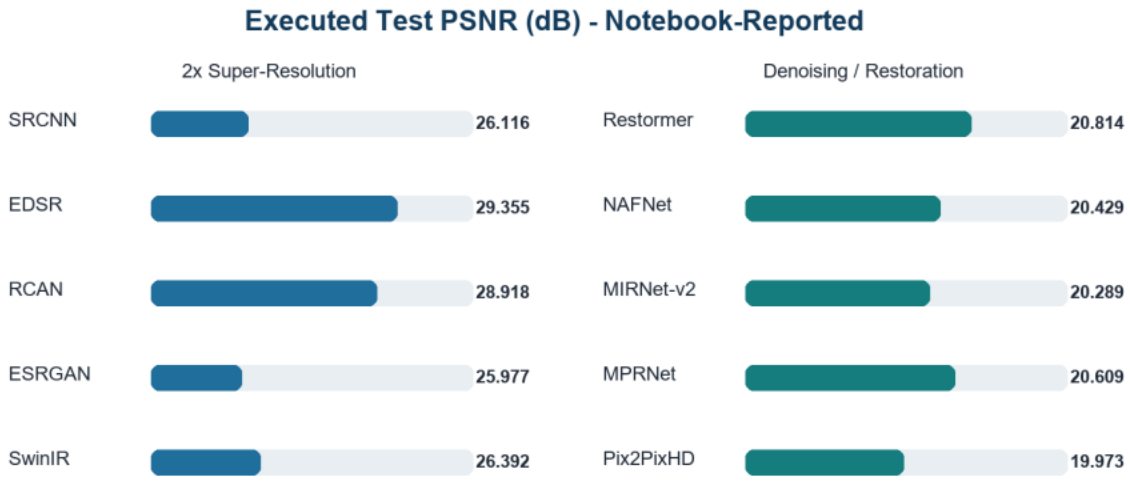



*Figure 1. Notebook-reported test PSNR. EDSR is the strongest reported SR fidelity result; cross-notebook metric standardization remains necessary.*

## 8.3 EDSR analysis

EDSR achieved the strongest reported SR result: 29.3548 dB PSNR and 0.981856 SSIM with 1.37 million parameters. Its validation curve rises steadily toward approximately 29.3 dB while loss converges smoothly, with no pronounced late-stage divergence. The notebook trained for 300 epochs using 96-pixel LR crops, batch size 8, AdamW at 2e-4, cosine decay, and validation every three epochs.

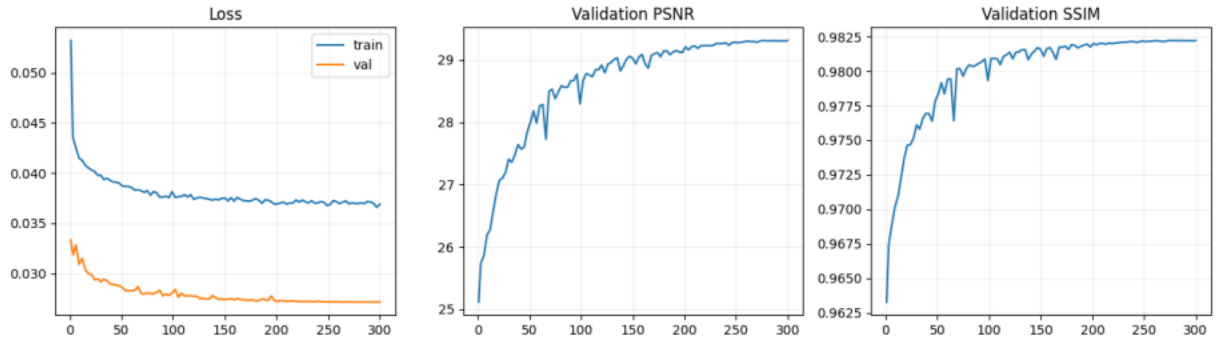



*Figure 2. EDSR training loss and validation fidelity across 300 epochs.*

This balance of accuracy, stability, latency, parameter count, and a direct PixelShuffle x2 reconstruction path justifies EDSR as the deployed SR component. It is also easier to audit than a GAN because its L1-driven output is less likely to invent semiconductor texture.

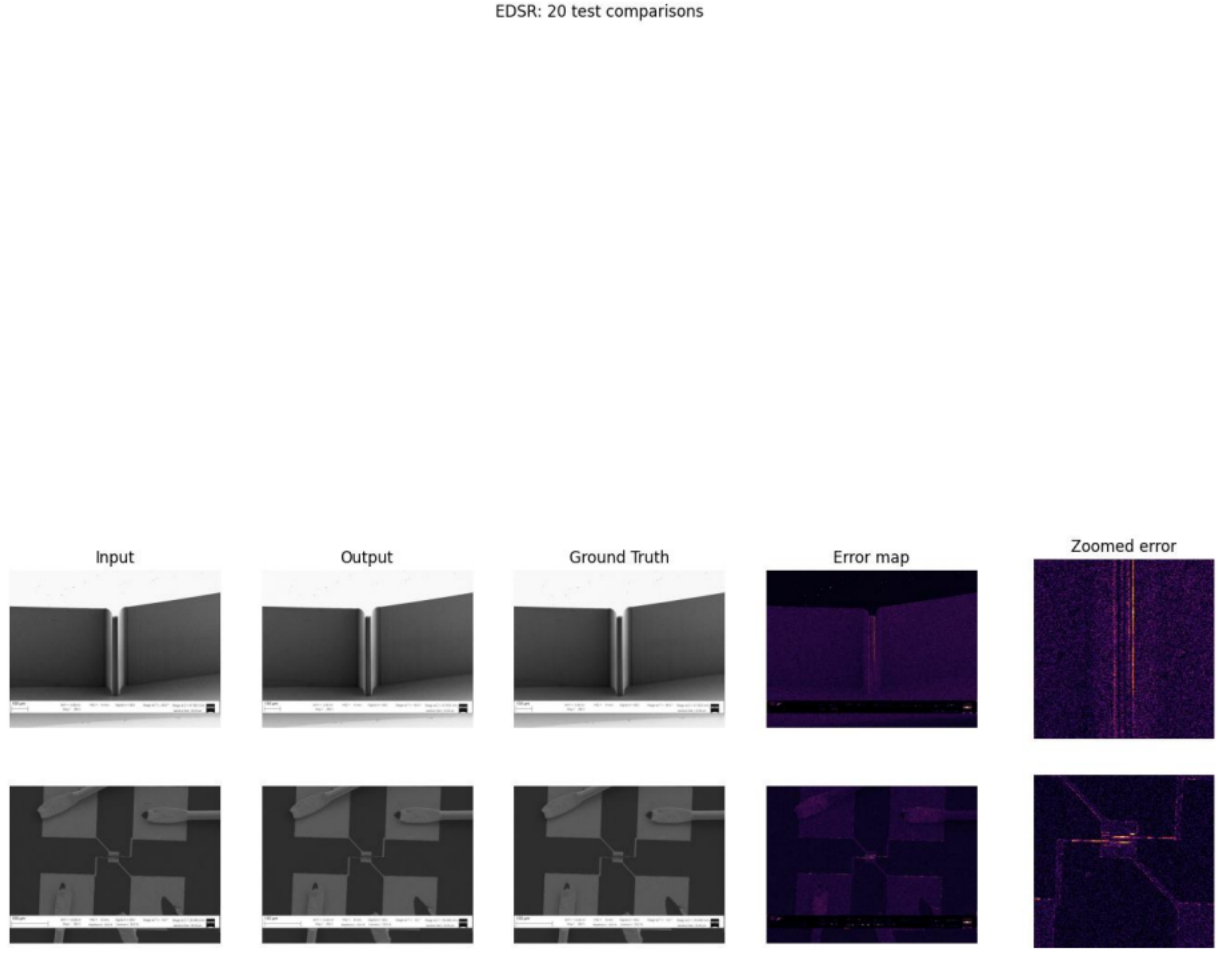



*Figure 3. Representative EDSR notebook test comparisons (cropped from the executed visualization output).*

## 8.4 RCAN, ESRGAN, SwinIR, and SRCNN interpretation

RCAN produced the second-highest reported PSNR (28.918 dB) and a smooth convergence curve through best epoch 159. Its channel-attention mechanism is technically well matched to repetitive SEM structures, but its 4.09 million parameters and approximately 472 ms/image test latency make it less attractive than EDSR for the desktop pipeline.

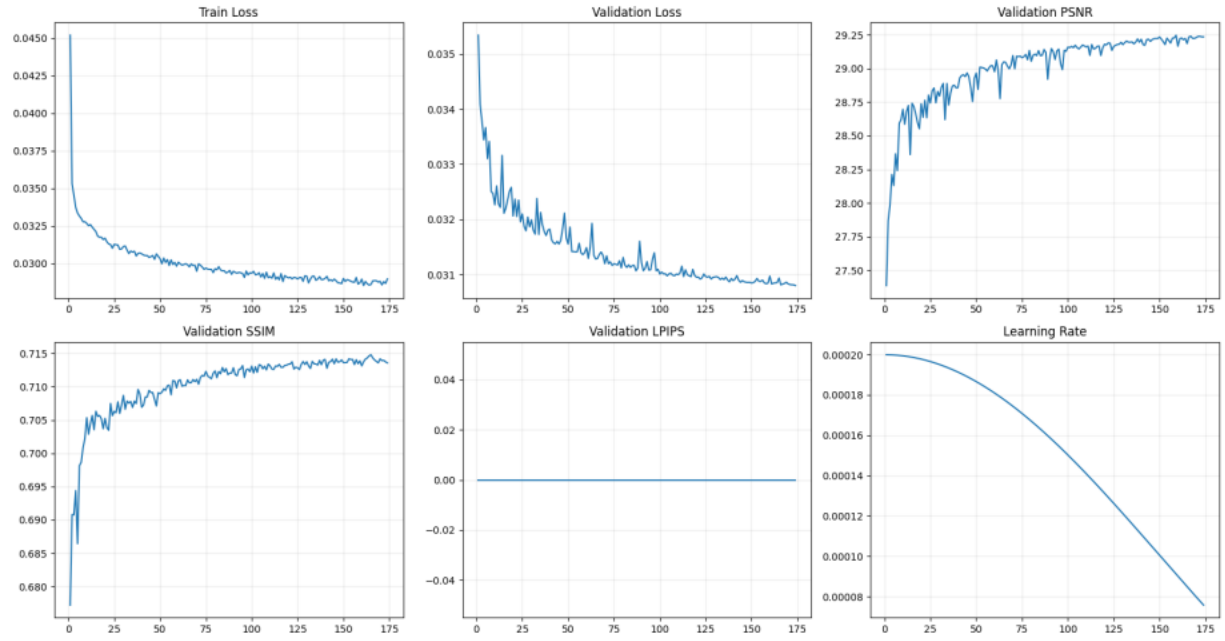



*Figure 4. RCAN convergence and cosine learning-rate schedule; validation LPIPS was recorded as a zero placeholder during training.*

ESRGAN illustrates the perception-distortion trade-off. Generator pretraining improved validation fidelity until epoch 50; switching to adversarial training sharply increased generator loss and reduced validation PSNR/SSIM. The final test LPIPS of 0.108 was better than RCAN's 0.143, but PSNR fell to 25.977 dB. For semiconductor inspection this is a critical warning: plausible GAN texture can look sharp while changing defect evidence.

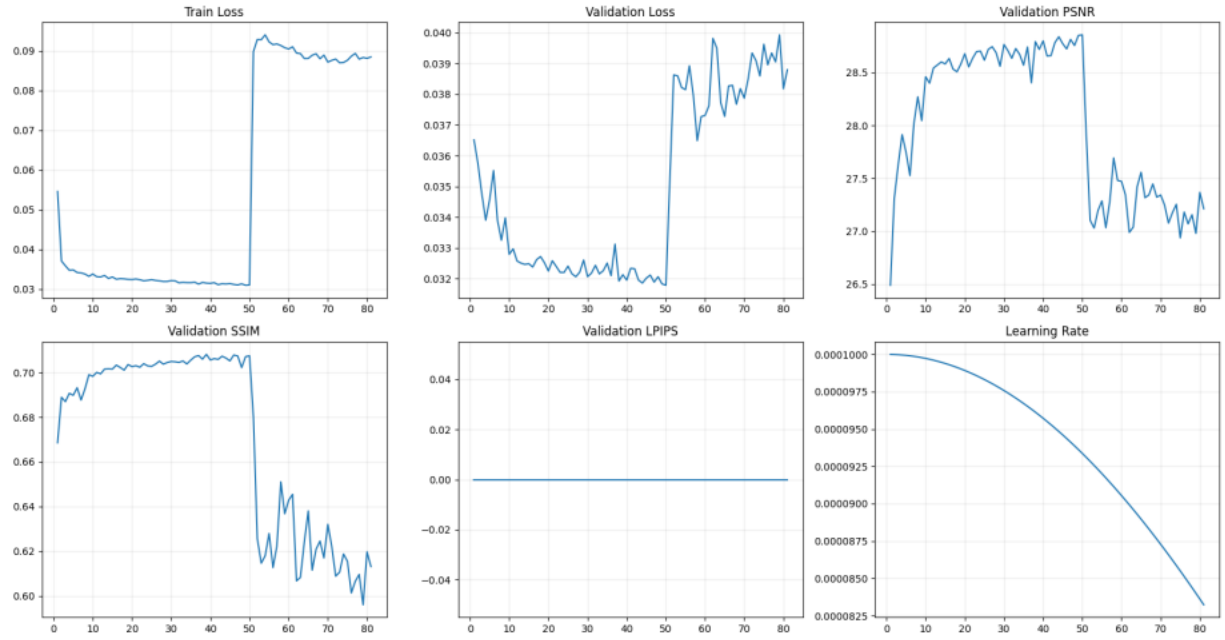



*Figure 5. ESRGAN stage transition at epoch 51 and the corresponding fidelity decline.*

SwinIR was deliberately optimized for Kaggle feasibility: 100 epochs, 1,024 sampled patches per epoch, 96-pixel crops, validation every five epochs on 64 images, and 256-pixel tiled inference. Its 25.657 dB best validation PSNR and 26.392 dB test PSNR demonstrate stable learning, but the compact configuration did not exceed EDSR.

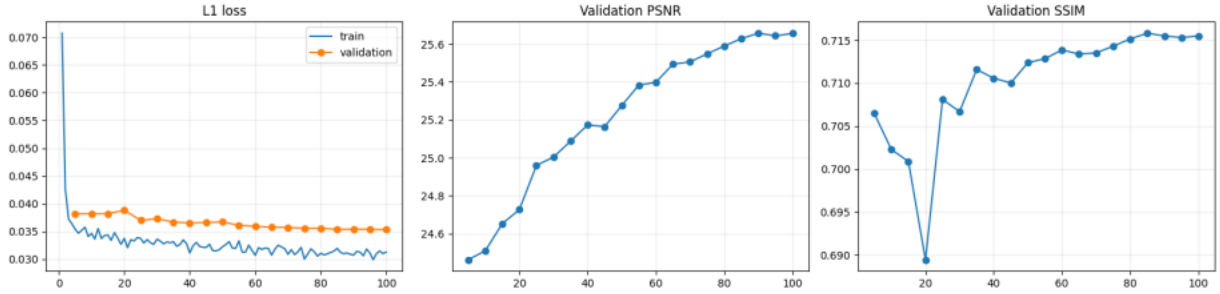



*Figure 6. Lightweight SwinIR learning curves with periodic subset validation.*

# 9. Denoising Models and Results

## 9.1 Implemented families

| Model | Architecture concept | Expected behavior |
|---|---|---|
| Restormer | Efficient attention and gated depthwise feed-forward blocks | Long-range artifact modeling |
| NAFNet | Activation-free residual restoration blocks | High-capacity stable fidelity |
| MIRNet-v2 | Multi-resolution branches and selective feature fusion | Fine/coarse structure integration |
| MPRNet | Progressive stages with supervised attention | Iterative refinement |
| Pix2PixHD | Conditional generator and patch discriminator | Perceptual realism with GAN risk |

## 9.2 Executed test results

| Model | PSNR dB | SSIM | Additional | Parameters | Latency |
|---|---|---|---|---|---|
| Restormer | 20.814 | 0.5583* | 0.4085 LPIPS | 325,249 | 1.5846 s |
| NAFNet | 20.429 | 0.8637 | 0.06484 MAE | 29,160,481 | 0.6131 s |
| MIRNet-v2 | 20.289 | 0.8590 | 0.06929 MAE | 1,485,281 | 0.6820 s |
| MPRNet | 20.609 | 0.5574* | 0.06637 MAE | 322,503 | 0.2426 s |
| Pix2PixHD | 19.973 | 0.8496 | 0.07085 MAE | 11,652,353 | 0.1870 s |

**The Restormer and MPRNet SSIM values are produced by the alternate notebook family. Their PSNR is credible within the reported run, but the large SSIM discrepancy relative to models with similar PSNR signals an evaluation-code inconsistency rather than a definitive structural-quality gap.*

## 9.3 Model-level observations

Restormer achieved the highest reported denoising PSNR at 20.814 dB and supplied the checkpoint selected for the application. The notebook best epoch was 96. The implementation is a compact adaptation rather than the paper's full four-level encoder-decoder, which explains the unusually low parameter count and should be stated whenever results are compared to published Restormer numbers.

NAFNet achieved the highest reported denoising SSIM within the common metric family (0.8637) and the lowest MAE (0.06484), but its 29.16 million parameters make it the largest denoising checkpoint. MIRNet-v2 delivered similar structural quality with only 1.49 million parameters. MPRNet provided the fastest non-GAN progressive

result at 0.243 seconds/image. Pix2PixHD was operationally fast but lowest in PSNR, consistent with adversarial objectives emphasizing appearance over exact recovery.

## 9.4 Training behavior and visual evidence

NAFNet's training loss decreases consistently over 100 epochs under cosine decay. Validation is performed periodically, and the best checkpoint is selected by PSNR. The final test table reports 20.429 dB, 0.863748 SSIM, 6.474 NIQE, 34.197 BRISQUE, 38.940 PIQE, 0.835 FSIM, and 6.431 entropy.

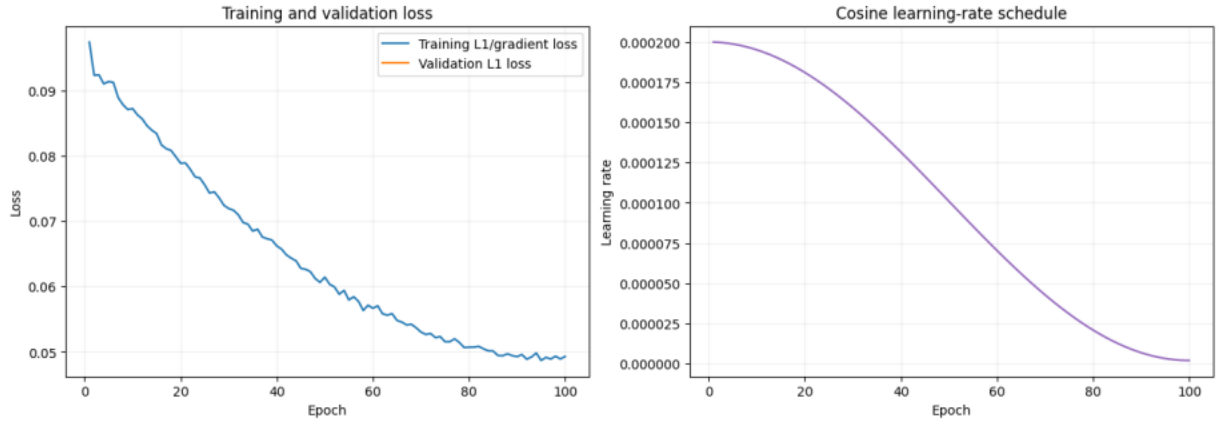



*Figure 7. NAFNet training loss and cosine learning-rate schedule.*

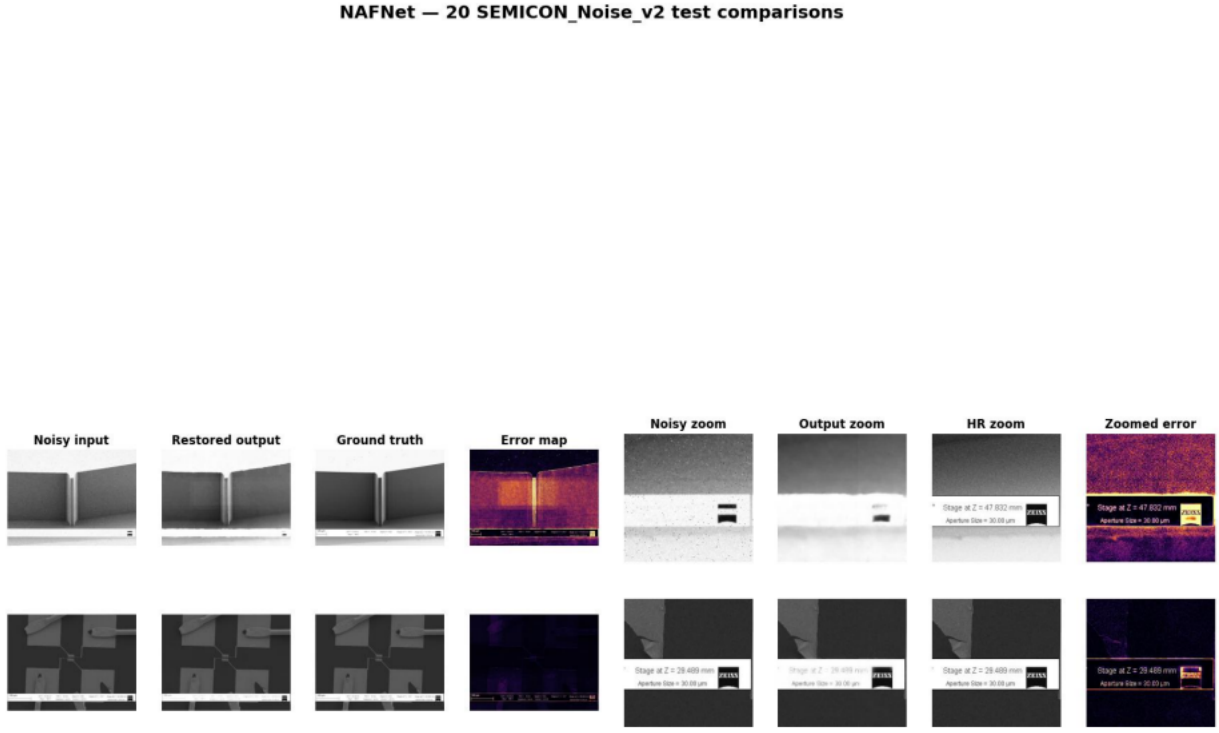



*Figure 8. Representative NAFNet notebook test comparisons and error maps (cropped from executed output).*

MIRNet-v2, MPRNet, and Pix2PixHD all complete full test evaluation with deferred no-reference IQA. The notebooks correctly avoid loading heavy NIQE/BRISQUE/PIQE/FSIM models during training, reducing memory pressure. Their full-image tiled inference is appropriate for 1024x768 inputs on T4/P100 hardware.

# 10. Cross-Model Interpretation

## 10.1 What the results support

Within the supplied evidence, EDSR is the clearest SR winner: it leads PSNR by 0.44 dB over RCAN while using one-third as many parameters and substantially lower latency. The result is also supported by smooth 300-epoch convergence and a high SSIM under the same notebook family as SRCNN. Restormer leads denoising PSNR by a smaller margin, while NAFNet and MIRNet-v2 lead the common-family SSIM results. The evidence therefore supports EDSR strongly and supports Restormer conditionally, with a standardized denoising rerun recommended.

## 10.2 Fairness limitations

- Epoch budgets differ: SR notebooks use 100 or 300 epochs; denoising uses 100 epochs.
- Validation cadence ranges from every epoch to every 3, 5, 6, or 8 epochs and may use full or subset validation.
- Crop sizes, samples per epoch, model capacity, tiling, and loss definitions differ by architecture.
- Two metric code families produce non-comparable SSIM scales for similar PSNR values.
- LPIPS is absent from some final tables and zero during some validation histories.
- No-reference IQA models use natural-image statistics and may penalize authentic SEM texture.

## 10.3 Recommended controlled rerun

7. Freeze one metric module, intensity range, border policy, tiling overlap, and test manifest.
8. Evaluate all best checkpoints with the same batch-size-1 evaluator on all 417 test images.
9. Report aggregate, single, combined, and per-degradation denoising metrics.
10. Add bicubic-input baseline metrics to quantify actual model gain.
11. Include region-of-interest metrics excluding annotation bars as a secondary analysis.
12. Repeat inference timing after warm-up on one named GPU and on the final CPU deployment system.

# 11. Application Architecture and User Workflow

## 11.1 Fixed inference pipeline

**DEPLOYED SEQUENCE** Original SEM image -> Restormer denoising/restoration -> EDSR x2 super-resolution -> final output

The final PyQt6 application intentionally removes model selectors and configuration panels. A sample number must be entered before capture or image loading is enabled. The interface then displays the original, restored, and super-resolved images simultaneously. This fixed workflow reduces operator error and makes the demonstration easy to understand during judging.

## 11.2 Model implementation

The application implements a compact one-channel Restormer-style network with 48 channels and 12 attention blocks, followed by a 64-channel, 16-block EDSR x2 network using PixelShuffle. Both checkpoints are loaded with strict state-dictionary matching. Inference converts to grayscale [0,1], uses overlap-tiled processing to control memory, clamps output to the valid image range, and converts back to 8-bit grayscale for display and saving.

## 11.3 Operator-facing metrics

| Panel | Displayed information | Interpretation |
|---|---|---|
| Original | Resolution | Input geometry only |
| Restored | BRISQUE, PIQE, estimated noise, entropy | Compact no-reference diagnostics |
| Super resolution | Resolution, NIQE, sharpness, edge density | Output scale and texture indicators |

**IMPORTANT LIMITATION** The desktop BRISQUE, PIQE, and NIQE values are lightweight offline heuristics implemented in app.py, not the canonical pyiqa models used in final notebook evaluation. They should be labeled as operational indicators, not benchmark scores.

## 11.4 User-interface result

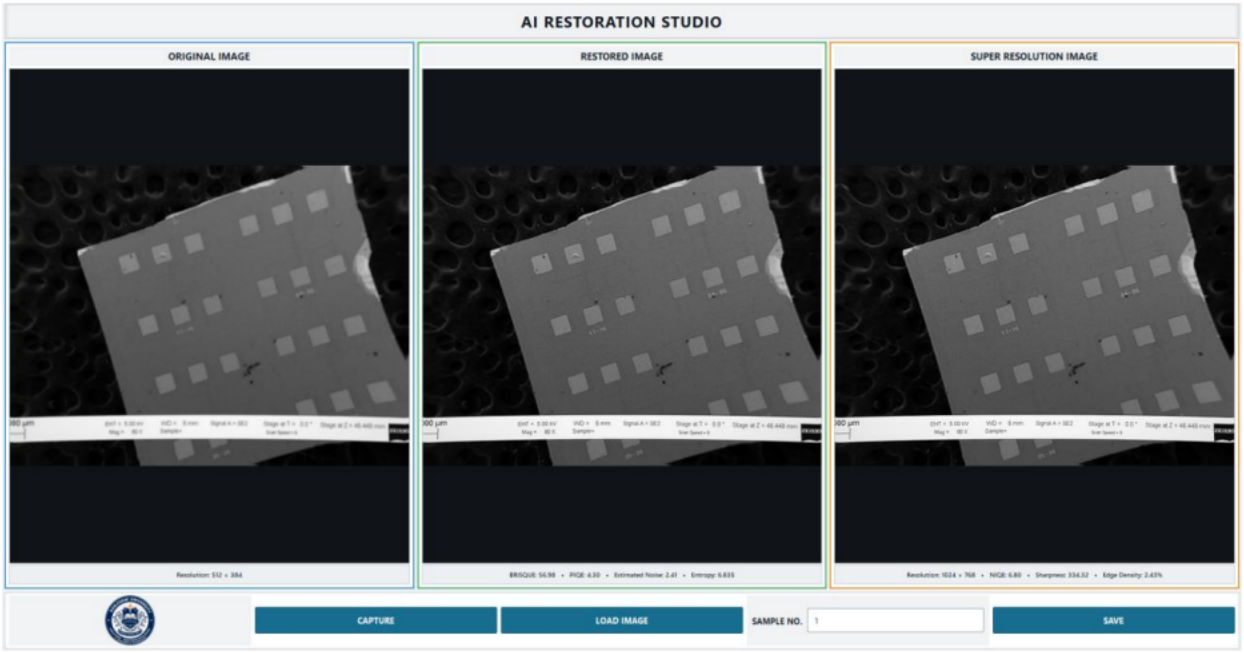



*Figure 9. Final native desktop interface showing the three-stage SEM restoration workflow.*

The application uses the full desktop width and prioritizes image viewing. Blue, green, and orange borders distinguish input, restoration, and super-resolution stages. Metrics are compact status strips rather than dashboard cards, preserving the microscopy-workstation character. The bottom toolbar provides institutional branding, capture, loading, sample entry, and save actions without exposing model complexity to the operator.

## 11.5 Functional scope

- Load JPEG, PNG, TIFF, or camera input.
- Require and display an operator-entered sample number.
- Execute both networks in a background thread so the interface remains responsive.
- Show all processing stages simultaneously and save the final SR image.
- Operate offline without APIs, browser services, or cloud inference.

# 12. Windows Packaging and Portability

## 12.1 Standalone packaging

The application was packaged as a 64-bit PyInstaller one-file executable and an Inno Setup installer. The audited executable size is 204,973,830 bytes (195.48 MiB), with SHA-256 225E4321BE941260A47803708A90EC786503A6AE9C0F7B7B1972FE5C115AD34B. Its CArchive contains 1,540 entries.

| Embedded component | Audit result |
|---|---|
| Restormer checkpoint | PASS - weights\best_restormer.zip |
| EDSR checkpoint | PASS - weights\best_edsr.pth |
| Python 3.12 runtime | PASS |
| Torch CPU runtime | PASS |
| Qt Core/GUI/Widgets and qwindows plugin | PASS |
| OpenCV extension and FFmpeg DLL | PASS |
| NumPy/Pillow native extensions | PASS |
| VC++ runtime DLLs | PASS |
| Unresolved non-system PE dependencies | None |

## 12.2 Clean-machine requirements

No Python, CUDA, PyTorch, PyQt6, OpenCV, model file, or internet connection is required. The remaining dependencies are normal operating-system conditions: 64-bit Windows 10/11, a writable TEMP directory for one-file extraction, standard Windows DLLs, Windows Media Foundation for camera support, and permission from endpoint security to execute unsigned software.

## 12.3 Distribution issue

Sharing the unsigned EXE through WhatsApp caused a security warning on another PC. Static packaging audit did not find missing application dependencies. The warning is best explained by Mark-of-the-Web, SmartScreen reputation, antivirus inspection of extracted Torch/Qt DLLs, or enterprise AppLocker/WDAC policy. A trusted code-signing certificate is the professional release solution; repackaging cannot legitimately bypass security policy.

**DEPLOYMENT CONCLUSION** The build is technically self-contained, but universal warning-free execution cannot be guaranteed for an unsigned Windows executable. Release signing and a clean-VM acceptance test are required before public distribution.

# 13. Engineering Challenges and Resolutions

| Challenge | Engineering response | Outcome |
|---|---|---|
| Dataset multiplication exhausted Kaggle storage | Replaced patch pre-generation with one pair per source and later separated SR/noise tasks | Lower storage and clearer protocol |
| Weak single degradations appeared visually clean | Audited PSNR/SSIM/MAE, then strengthened charging, Gaussian, speckle, gamma, and vignetting | More meaningful restoration signal |
| Strong-update manifest preserved old parameters | Detected and documented as provenance debt | Requires corrected sidecar/manifest before freeze |
| P100 incompatible with installed CUDA build | Adjusted environment expectations and emphasized T4/P100-compatible builds | Reduced runtime failure risk |
| Large models caused CUDA OOM | Reduced batches, used aligned patches, AMP, gradient checkpointing, tiling, and deferred IQA | Kaggle-feasible execution |
| Validation dominated runtime | Reduced cadence and used deterministic subsets for expensive models | Shorter benchmark cycle |
| GAN stage reduced fidelity | Retained best-validation checkpoints and treated GANs as perceptual secondary models | Controlled hallucination risk |
| PyQt/PyInstaller dependency failures | Pinned runtime packages and explicitly bundled OpenCV, Torch source, Qt plugins, weights, assets and VC DLLs | Self-contained EXE |
| Security warning after sharing | Verified archive completeness and identified signing/OS policy as external distribution concern | Clear deployment remediation |

These iterations are an important part of the project contribution. They show that restoration research is inseparable from data provenance, memory-aware experimentation, consistent evaluation, and deployment engineering.

# 14. Limitations, Ethics, and Risk Control

## 14.1 Scientific limitations

| Risk | Potential consequence | Mitigation |
|---|---|---|
| Synthetic-to-real gap | Weak transfer to official target degradations | Fine-tune and select on official-domain validation |
| Duplicate groups | Optimistic validation/test estimates | Group exact and near duplicates in future split |
| Fixed resizing | Altered physical scale | Disclose preprocessing; use scale-aware crops |
| Annotation bars | Shortcut learning and metric distortion | Add content-ROI ablation or masks |
| All-11 combined degradation | Unrealistic severity/order | Create calibrated severity mixtures in future work |
| Stale per-image parameters | Incomplete reproducibility | Publish repaired manifest/update log |
| Natural-image IQA | Misleading SEM quality judgment | Treat as supplementary; use expert review |
| GAN hallucination | Invented edges or texture | Use GANs only with fidelity/uncertainty checks |

## 14.2 Responsible interpretation

Restored images are algorithmic estimates, not new physical measurements. A visually sharpened edge must not be interpreted as proof of a defect or dimensional change without confirmation from the original image, acquisition metadata, or repeat measurement. This is especially important for GAN outputs, whose objective can favor plausible texture rather than exact reconstruction.

## 14.3 Dataset and licensing governance

The final publication should retain NFFA-Europe attribution and source licensing terms, document transformation and redistribution rights, publish immutable archive hashes, and maintain a changelog. The supplied evidence does not include the authoritative license text; therefore licensing compliance must be confirmed from the source dataset before redistribution or commercial use.

# 15. Research Strategy and Recommended Final Pipeline

## 15.1 Recommended sequence

**PRIMARY SUBMISSION PIPELINE** Degraded SEM image -> Restormer or best standardized denoiser -> EDSR x2 -> intensity-safe final output

Denoising before upscaling prevents stochastic noise and charging streaks from being spatially magnified. EDSR then reconstructs the bicubic x2 target with the strongest reported fidelity. The reverse order should still be evaluated as an ablation because the official degradation sequence may differ.

## 15.2 Model decision

| Decision | Recommended model | Rationale |
|---|---|---|
| SR baseline/final candidate | EDSR | Highest PSNR/SSIM, stable convergence, practical latency |
| SR alternate | RCAN | Strong fidelity and channel attention; heavier |
| Perceptual SR | ESRGAN | Use only if LPIPS/visual quality is rewarded and hallucination is controlled |
| Denoising deployed | Restormer | Highest reported PSNR and existing integrated checkpoint |
| Denoising challenger | NAFNet | Strong common-family SSIM/MAE; larger model |
| Efficient denoising | MPRNet compact | Fast progressive inference; metric rerun needed |

## 15.3 Final competition work plan

13. Repair Noise-v2 strong-degradation metadata and publish hashes.
14. Run one standardized evaluator over all supplied best checkpoints.
15. Establish bicubic/no-processing baselines and content-ROI metrics.
16. Fine-tune Restormer/NAFNet and EDSR/RCAN on official **target** data.
17. Evaluate denoise->SR, SR->denoise, and joint fine-tuning as controlled ablations.
18. Select the final checkpoint using official validation PSNR/SSIM/LPIPS plus expert defect-preservation review.
19. Measure end-to-end latency on the judging machine and sign the Windows release.

# 16. Conclusions and Final Assessment

The project demonstrates a credible end-to-end semiconductor image restoration program. It creates and audits two complementary paired datasets, explores ten model configurations across major restoration families, records quantitative and visual outputs, selects practical models, and delivers an offline native desktop workstation. The strongest dataset-engineering feature is the shared deterministic source-level split; the strongest model evidence is EDSR's 29.355 dB test PSNR with stable 300-epoch convergence; and the strongest systems contribution is the fully embedded Restormer/EDSR Windows package.

The work is suitable for **Research** evaluation and academic presentation provided its limitations are stated plainly. The synthetic dataset is an external pretraining and controlled research resource, not the official target benchmark. The duplicate groups and stale strong-noise metadata are real reproducibility risks. The benchmark tables are evidence-bearing but not yet a perfectly controlled leaderboard because of metric and validation differences between notebooks.

**FINAL VERDICT** APPROVED WITH DOCUMENTED TECHNICAL LIMITATIONS. The project has substantial engineering depth, a coherent experimental lifecycle, and a working deployment. Priority before final submission is standardized checkpoint evaluation, official-domain fine-tuning, metadata repair, and signed application distribution.

## 16.1 Principal achievements

- 8,316 valid paired samples across two task-specific datasets.
- Exact 2x geometry and complete same-resolution denoising pairs.
- Ten executed Kaggle notebooks with checkpoint, curve, visualization, and metric outputs.
- Evidence-based choice of EDSR and Restormer for a fixed operator workflow.
- Professional one-screen PyQt6 application with offline inference.
- Self-contained 195.48 MiB executable with verified embedded models and dependencies.
- Transparent documentation of dataset, evaluation, metadata, and deployment risks.

# Appendix A. Detailed Notebook Inventory

| Notebook | Task | Executed result | Notable configuration |
|---|---|---|---|
| srcnnmodel.ipynb | SR | 26.116 dB | 300 epochs; crop 128; batch 16 |
| edsrmodel.ipynb | SR | 29.355 dB | 300 epochs; crop 96; batch 8 |
| rcanmodel (1).ipynb | SR | 28.918 dB | Early stop 174; best 159 |
| esrganmodel (1).ipynb | SR GAN | 25.977 dB | 50-epoch pretrain; early stop 81 |
| swinir...ipynb | SR | 26.392 dB | 100 epochs; 1,024 patches/epoch |
| restormer.ipynb | Denoise | 20.814 dB | 100 epochs; best 96; batch 4 x accum 2 |
| nafnet.ipynb | Denoise | 20.429 dB | 100 epochs; crop 192; batch 8 |
| mir-model.ipynb | Denoise | 20.289 dB | 100 epochs; crop 192; batch 8 |
| MPRNet.ipynb | Denoise | 20.609 dB | 100 epochs; crop 192; batch 2 |
| pix2-1.ipynb | Denoise GAN | 19.973 dB | 100 epochs; crop 256; batch 4 |

All ten notebooks contained executed code cells, embedded plots, and no stored exception traceback. Some final IQA progress bars show 0% snapshots in captured output, but final result tables were present.

# Appendix B. Checkpoint Inventory

The weight archive contained nine PyTorch ZIP-format checkpoints: four denoising entries and five SR entries. Archive inspection confirmed PyTorch serialization structures (data.pkl, tensor storage entries, version, byte order). Filenames and state keys identify Restormer, ESRGAN, RCAN, SwinIR, and EDSR-related checkpoints; generic best/best_checkpoint files require notebook-to-hash mapping before release.

| Task group | Checkpoint files | Verification level |
|---|---|---|
| Denoising | best (2), two best_checkpoint copies, best_restormer (2) | Archive valid; Restormer app load verified |
| Super-resolution | two best_checkpoint files, best_esrgan, best_rcan, SwinIR | Archive valid; app EDSR checkpoint separately embedded/verified |

**ARTIFACT GAP** The archive lacks a signed registry mapping every generic checkpoint filename to notebook, architecture configuration, best epoch, dataset version, and metric row. Add a weights_manifest.json with SHA-256 hashes before archival.

# Appendix C. Reproducibility Checklist

- [ ] Freeze Kaggle dataset version and archive SHA-256.
- [ ] Publish shared split ledger and both manifests.
- [ ] Publish corrected strong-noise per-image parameters.
- [ ] Record library versions for data generation and training.
- [ ] Map every checkpoint hash to architecture and notebook.
- [ ] Use one test evaluator and document SSIM window/border policy.
- [ ] Record seed, crop, batch, accumulation, optimizer, scheduler, validation cadence, and early stopping.
- [ ] Retain per-image metric CSVs and selected visual comparisons.
- [ ] Measure latency after warm-up on a named device.
- [ ] Confirm NFFA-Europe license and attribution obligations.
- [ ] Sign the final Windows executable.
- [ ] Validate the release in a clean Windows 10/11 VM with no Python installed.

# Appendix D. Metric Definitions

PSNR = 10 log10(MAX^2/MSE), where MAX is 1 for normalized images. MAE is the mean absolute pixel difference; MSE is the mean squared pixel difference; RMSE is its square root. SSIM compares local luminance, contrast, and structure. LPIPS uses deep features to approximate perceptual distance. FSIM compares phase congruency and gradient magnitude. NIQE, BRISQUE, and PIQE estimate no-reference quality from natural-image statistics. Entropy summarizes histogram uncertainty. Sharpness in the application is variance of the Laplacian; edge density is the fraction of Canny edge pixels.

For SEM, natural-image metrics are supplementary. A low NIQE or BRISQUE score does not prove that a restored edge is physically correct, and high entropy can result from either useful detail or noise.

# Appendix E. References and Source Links

1. Project dataset: SEMICON v2 Dual Dataset. https://www.kaggle.com/datasets/omkadam05/semicon-v2-dual-dataset

2. Lim et al. Enhanced Deep Residual Networks for Single Image Super-Resolution. CVPR Workshops, 2017. https://openaccess.thecvf.com/content_cvpr_2017_workshops/w12/html/Lim_Enhanced_Deep_Residual_CVPR_2017_paper.html

3. Zhang et al. Image Super-Resolution Using Very Deep Residual Channel Attention Networks. ECCV, 2018. https://openaccess.thecvf.com/content_ECCV_2018/html/Yulun_Zhang_Image_Super-Resolution_Using_ECCV_2018_paper.html

4. Wang et al. ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks. ECCV Workshops, 2018. https://openaccess.thecvf.com/content_eccv_2018_workshops/w25/html/Wang_ESRGAN_Enhanced_Super-Resolution_Generative_Adversarial_Networks_ECCVW_2018_paper.html

5. Liang et al. SwinIR: Image Restoration Using Swin Transformer. ICCV Workshops, 2021. https://openaccess.thecvf.com/content/ICCV2021W/AIM/html/Liang_SwinIR_Image_Restoration_Using_Swin_Transformer_ICCVW_2021_paper.html

6. Zamir et al. Multi-Stage Progressive Image Restoration. CVPR, 2021. https://openaccess.thecvf.com/content/CVPR2021/html/Zamir_Multi-Stage_Progressive_Image_Restoration_CVPR_2021_paper.html

7. Zamir et al. Restormer: Efficient Transformer for High-Resolution Image Restoration. CVPR, 2022. https://openaccess.thecvf.com/content/CVPR2022/papers/Zamir_Restormer_Efficient_Transformer_for_High-Resolution_Image_Restoration_CVPR_2022_paper

8. Chen et al. Simple Baselines for Image Restoration (NAFNet). ECCV, 2022. https://arxiv.org/abs/2204.04676

9. Zamir et al. Learning Enriched Features for Real Image Restoration and Enhancement (MIRNet). ECCV, 2020. https://arxiv.org/abs/2003.06792

10. Isola et al. Image-to-Image Translation with Conditional Adversarial Networks. CVPR, 2017. https://openaccess.thecvf.com/content_cvpr_2017/html/Isola_Image-To-Image_Translation_With_CVPR_2017_paper.html

# Appendix F. Evidence Gaps to Close

- Author/team names, mentor name, department, and formal submission metadata were not present in the supplied artifacts.
- The authoritative NFFA-Europe license text and exact source citation were not included.
- Official target validation/test results were not supplied; all quantitative results here are external synthetic-benchmark results.
- A corrected strong-noise manifest/update log was not included.
- A single standardized metric rerun across all checkpoints was not supplied.
- The application was statically audited and tested in isolation on the build machine, but a signed clean-VM acceptance certificate was not supplied.

**ARCHIVAL STATUS** The project is technically mature enough for submission and demonstration. Closing the evidence gaps above would elevate it from a strong Research artifact to a fully reproducible research release.